## Highlights

A signal-to-drift ratio for quantifying winding-fault detectability and selecting condition monitoring across three wind-turbine generator types

Hüseyin Tayyer Canseven

- A signal-to-drift ratio quantifies fault detectability without a model or detector
- Phasor residuals remove the faulted-phase dependence of negative-sequence ratios
- Signatures relocate across voltage, controller and field current by machine and drive
- A decision table selects the cheapest adequate sensor suite with bootstrap confidence
- Detectors transfer across generators once healthy-referenced (AUC 0.85 to 0.98)